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Traffic Prediction using Machine Learning

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Abstract. The paper deals with traffic prediction that can be done in intelligent transportation systems which involve the prediction between previous year's data set and the recent year data which ultimately provides the accuracy and mean square error. This prediction will be helpful for the people who are in need to check the immediate traffic state. The traffic data is predicated on a basis of 1 hour time gap. Live statistics of the traffic is analyzed from this prediction. So this will be easier to analyze when the user is on driving too. The system compares the data of all roads and determines the most populated roads of the city. I propose the regression model in order to predict the traffic using machine learning by importing Sklearn, Keras and Tensorflow libraries

Keywords: Traffic, Regression, Intelligent Transport System (ITS) , Machine learning, Prediction

1 Introduction

Machine Learning (ML) is one of the most important and popular emerging branches these days as it is a part of Artificial Intelligence (AI). In recent times, machine learning becomes an essential and upcoming research area for transportation engineering, especially in traffic prediction. Traffic congestion affects the country's economy directly or indirectly by its means. Traffic congestion also takes people's valuable time, cost of fuel every single day. As traffic congestion is a major problem for all classes in society, there has to be a small-scale traffic prediction for the people's sake of living their lives without frustration or tension. For ensuring the country's economic growth, the road user's ease is required in the first place. This is possible only when the traffic flow is smooth. To deal with this, Traffic prediction is needed so that we can estimate or predict the future traffic to some extent.

In addition to the country's economy, pollution can also be reduced. The government is also investing in the intelligent transportation system (ITS) to solve these issues. The plot of this research paper is to find different machine learning algorithms and speculating the models by utilizing python3. The goal of traffic

flow prediction is to predict the traffic to the users as soon as possible. Nowadays the traffic becomes really hectic and this cannot be determined by the people when they are on roads. So, this research can be helpful to predict traffic. Machine learning is usually done using anaconda software but in this paper, I have used the python program using command prompt window which is much easier than the usual way of predicting the data [16]. In summary, the constructs of this paper consist of ten major sections. These are: Introduction, Purpose of Traffic Prediction, Problem Statement, Related Work, Overview, Methodology, Software Implementation and Conclusion with Future work.

1.1 Purpose of statement

Many reports of the traffic data are of actual time but it is not favorable and accessible to many users as we need to have prior decision in which route we need to travel. For example, During working days, we need to have daily traffic information or at times we need hourly traffic information but then the traffic congestion occurs; for solving this issue the user need to have actual time traffic prediction. Many factors are responsible for the traffic congestion. This can be predicted by taking two datasets; one with the past year and one with the recent year's data set. If traffic is so heavy then the traffic can be predicted by referring the same time in the past year's data set and analyzing how congested the traffic would be. With the increasing cost of the fuel, the traffic congestion changes drastically. The goal of this prediction is to provide real-time gridlock and snarl up information. The traffic on the city becomes complex and are out of control these days, so such kind of systems are not sufficient for prediction. Therefore, research on traffic flow prediction plays a major role in ITS. [8]

1.2 Problem Statement

To overcome the problem of traffic congestion, the traffic prediction using machine learning which contains regression model and libraries like pandas, os, numpy, matplotlib.pyplot are used to predict the traffic. This has to be implemented so that the traffic congestion is controlled and can be accessed easily. Users can collect the traffic information of the traffic flow and can also check the congestion flow from the start of the day till the end of the day with the time span of one hour data. In this way, Users can know the weather conditions of the roads that they would probably opt to take. This also tells the accuracy of the traffic by comparing their mean square errors of the past year's data and the recent year's data. Users can also know how many vehicles are traveling on average by the traffic prediction.

2 Related Works

A person developed the LSTM based prediction models by using machine learning approaches, which involve structure designing or network training designing and prediction. The next goal is to deal with prediction errors that may occur during the prediction process with deep learning methods. This method is applied to big data that has collected from the performance measurement system. Then the experiments show that the LSTM model performed well. Vlahogianni implemented a feeding device knowledge with an ANN (Artificial neural network), but some researchers implemented a completely different spatial associated temporal holdup to the setup [1],[9],[10],[17]. Some researchers worked on the traffic prediction using the neural networks and hybrid by various techniques. Traffic patterns have been identified by grouping them and the traffic flow have been identified by the neural networks. On other hand studies conducts two kinds of prediction models: Models which are not available in online are accessed through on historical information that can be trained when important changes appear in the system as like changes and updating the whole framework, Online models are measured by historical data and the system gets updated using usual transport condition achieved through v2v/v2i communication. [2],[7]

Many researchers say that LSTM is more capable compared to shallow machine learning models. Machine learning approaches big data for the prediction problems. With the lots of the data, the real time traffic flow can be predicted. LSTM is based on the deep learning. RNN may be used to communicate the prior information to the present situation. [3],[6],[20].

3 Overview

In traffic congestion forecasting there are data collection and prediction model.

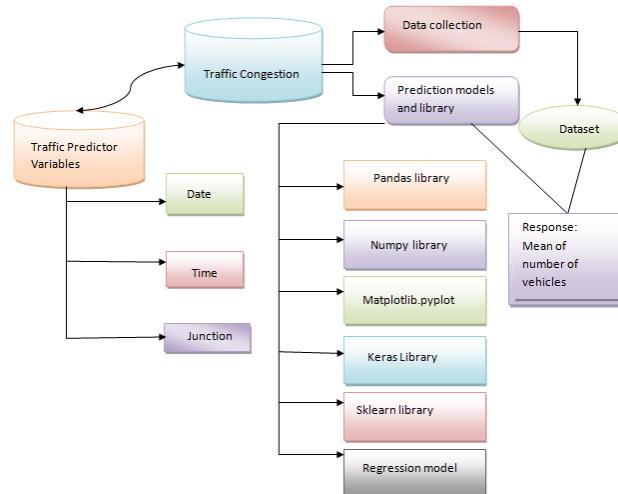


Figure 1. Outline of the traffic prediction in this paper

The methodology has to be done correctly so that there won't be any flaws while predicting. After data collection, the vital role is the data processing which is to train and test the datasets that is taken as the input. After processing the data, the validation of the model is done by using necessary models. Figure 1 highlights the outline of traffic prediction using machine learning.

4 Methodology

Many researchers have been used various discussed approaches. This paper contains the technique of predicting the traffic using regression model using various libraries like Pandas, Numpy, OS, Matplotlib.pyplot, Keras and Sklearn.

4.1 Data set

Traffic congestion is raising a lot these days. Factors like expanding urban populations, uncoordinated traffic signal timing and a lack of real-time data. The effect of the traffic congestion is very huge these days. Data collected in this paper are from the Kaggle website for the implementations of machine learning algorithms using python3 to show outputs in the traffic prediction. Two datasets are collected in which one is the 2015's traffic data which comprises of date, time, and number of vehicles, junction and the rest one is the 2017's traffic data with the same details so as to compare easily without any misconception. The unwanted data has been deleted by pre-processing the data aggregated from 1 to 24 hours time interval to calculate traffic flow prediction with each 1 hour interval. [11]

4.2 Regression model

Regressor model analysis could even be a mathematical technique for resolving the connection in the middle of one dependent (criterion) variable and one or more independent (predictor) variables.

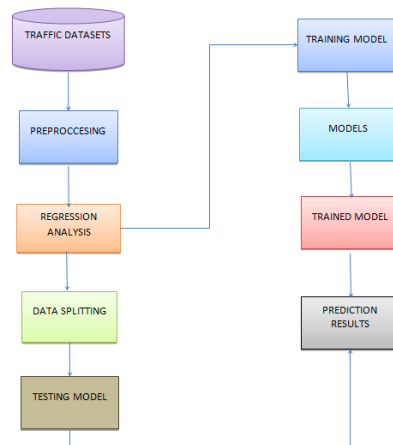


Figure 2. Regression model of traffic prediction in this paper

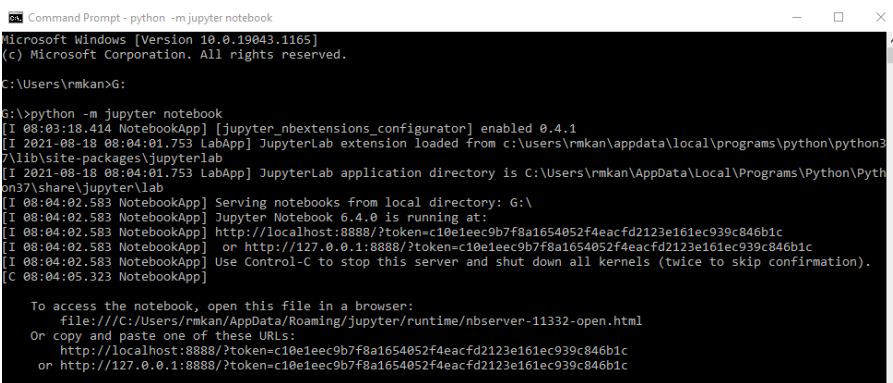
The evaluation yields a foretold value for the benchmark resulting from a sum of scalar vectors of the predictors. The accuracy is measured by computing mean square error. Thus obtaining the expected error from the observed value and also truth value which is equivalent to the standard deviation deployed within the statistical methods [4],[12].Figure 2 shows the Regression model for the Traffic Prediction.

JupyterLab is a browser based communal development. JupyterLab is a limber and which can construct and exhibit the user interface to support a far flung of metadata in machine learning [5].Python3 is the status quo environment where the code is implemented in Jupyter notebook. This can be accesses/installed using command prompt. This is done in order to get the access from the local drive. So the Jupyter notebook is installed through command prompt and then a local host is created. The file is accessed though this host and the prediction are done using various libraries and models in the python environment.

5 Software Implementation

5.1 Simulation

The command prompt is the local host in this paper to initialize the jupyter notebook.



```

Microsoft Windows [Version 10.0.19043.1165]
(c) Microsoft Corporation. All rights reserved.

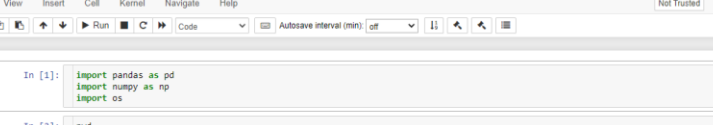
C:\Users\rmkan>G:
G:\>python -m jupyter notebook
[I 08:03:18.414 NotebookApp] [jupyter_nbextensions_configurator] enabled 0.4.1
[I 2021-08-18 08:04:01.753 LabApp] JupyterLab extension loaded from c:\users\rmkan\appdata\local\programs\python\python37\lib\site-packages\jupyterlab
[I 2021-08-18 08:04:01.753 LabApp] JupyterLab application directory is C:\Users\rmkan\AppData\Local\Programs\Python\Python37\Share\jupyterlab
[I 08:04:02.583 NotebookApp] Serving notebooks from local directory: G:\
[I 08:04:02.583 NotebookApp] Jupyter Notebook 6.4.0 is running at:
[I 08:04:02.583 NotebookApp] http://localhost:8888/?token=c10e1eec9b7f8a1654052f4eacfd2123e161ec939c846b1c
[I 08:04:02.583 NotebookApp] or http://127.0.0.1:8888/?token=c10e1eec9b7f8a1654052f4eacfd2123e161ec939c846b1c
[I 08:04:02.583 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 08:04:05.323 NotebookApp]

To access the notebook, open this file in a browser:
file:///C:/Users/rmkan/AppData/Roaming/jupyter/runtime/nbserver-11332-open.html
Or copy and paste one of these URLs:
http://localhost:8888/?token=c10e1eec9b7f8a1654052f4eacfd2123e161ec939c846b1c
or http://127.0.0.1:8888/?token=c10e1eec9b7f8a1654052f4eacfd2123e161ec939c846b1c

```

Figure 3. The figure signifies the initializing the jupyter notebook through command prompt

The local host contains the nbextensions which we modify to our convenience



The screenshot shows a Jupyter Notebook interface with the following content:

```
In [1]: import pandas as pd
import numpy as np
import os

In [2]: pwd

Out[2]: 'G:\\"

In [3]: pd.set_option('display.max_rows',10)

In [4]: def nl():
print('\n')

In [5]: df_train = pd.read_csv("G:\train_annotu8.csv",parse_dates = [0], infer_datetime_format = True)
df_train.head()

Out[5]:
```

	DateTime	Junction	Vehicles	ID
0	2015-11-01 00:00:00	1	15	20151101001
1	2015-11-01 01:00:00	1	13	20151101011
2	2015-11-01 02:00:00	1	10	20151101021
3	2015-11-01 03:00:00	1	7	20151101031
4	2015-11-01 04:00:00	1	9	20151101041

```
In [6]: df_test = pd.read_csv(r"G:\test_BdKkA\j.csv", parse_dates = [0], infer_datetime_format = True)
df_test.head()
```

```
Out[6]:
```

	DateTime	Junction	ID
0	2017-07-01 00:00:00	1	20170701001
1	2017-07-01 01:00:00	1	20170701011
2	2017-07-01 02:00:00	1	20170701021
3	2017-07-01 03:00:00	1	20170701031
4	2017-07-01 04:00:00	1	20170701041

Figure 6. The figure above denotes the result after running the particular command in the prediction process will be displayed under out command

The below mentioned are the libraries used for the prediction of traffic

```
import pandas as pd # library imported
import numpy as np # library imported
import os          # library imported
```

Table 1. The training model of the junction 1 for the prediction that is taken[19]

	DateTime	Junction	Vehicles	ID
0	2015-11-01 00:00:00	1	15	2015110 1001
1	2015-11-01 01:00:00	1	13	2015110 1011
2	2015-11-01 02:00:00	1	10	2015110 1021
3	2015-11-01 03:00:00	1	7	2015110 1031
4	2015-11-01 04:00:00	1	9	2015110 1041

Table 2. The testing model of the junction 1 for the prediction that is taken

DateTime	Junction	ID
2017-07-01 00:00:00	1	20170701001
2017-07-01 01:00:00	1	20170701011
2017-07-01 02:00:00	1	20170701021
2017-07-01 03:00:00	1	20170701031
2017-07-01 04:00:00	1	20170701041

The training model and testing model of Junction 1 is displayed using the following commands

```
time_targets = df_tmp.groupby([level_values(0)] + [pd. Grouper(freq =
'1M',level=-1))]['Vehicles'].sum()
```



```
train = df_train.pivot(index = 'DateTime', columns = 'Junction', values = 'Vehicles')
```

```
train = train.fillna(0)
```

Table 3. The pivot table of both the training and testing model without NaN values

Junction	1	2	3	4
DateTime				
2015-11-01 00:00:00	15.0	6.0	9.0	0.0
2015-11-01 01:00:00	13.0	6.0	7.0	0.0
2015-11-01 02:00:00	10.0	5.0	5.0	0.0
2015-11-01 03:00:00	7.0	6.0	1.0	0.0
2015-11-01 04:00:00	9.0	7.0	2.0	0.0
...	...	...	...	...
2017-06-30 19:00:00	105.0	34.0	33.0	11.0
2017-06-30 20:00:00	96.0	35.0	31.0	30.0
2017-06-30 21:00:00	90.0	31.0	28.0	16.0
2017-06-30 22:00:00	84.0	29.0	26.0	22.0
2017-06-30 23:00:00	78.0	27.0	39.0	12.0

The training model and testing model of Junction 1 is displayed using the following commands

```
Xy_train = gen_lag_features(train)
```

Table 4. The table of both the training and testing model of concatenated values

Junction	Junction	Junction	Junction	Junction	Junction	Junction	Junction
1(H-1)	2(H-1)	3(H-1)	4(H-1)	1(H)	2(H)	3(H)	4(H)
DateTime							
2015-11-01 01:00:00	15.0	6.0	9.0	0.0	13.0	6.0	7.0

2015-11-01 02:00:00	13.0	6.0	7.0	0.0	10.0	5.0	5.0
2015-11-01 03:00:00	10.0	5.0	5.0	0.0	7.0	6.0	1.0
2015-11-01 04:00:00	7.0	6.0	1.0	0.0	9.0	7.0	2.0
2015-11-01 05:00:00	9.0	7.0	2.0	0.0	6.0	2.0	2.0
...	...	...	...	...	...	...	...
2017-06-30 19:00:00	95.0	34.0	38.0	17.0	105.0	34.0	33.0
2017-06-30 20:00:00	105.0	34.0	33.0	11.0	96.0	35.0	31.0
2017-06-30 21:00:00	96.0	35.0	31.0	30.0	90.0	31.0	28.0
2017-06-30 22:00:00	90.0	31.0	28.0	16.0	84.0	29.0	26.0
2017-06-30 23:00:00	84.0	29.0	26.0	22.0	78.0	27.0	39.0

Command here used:

`Xy_train[Xy_train.columns]=scaler.fit_transform(Xy_train[Xy_train.columns])`

Table 5. The table contains the training model's results where the scaling of the training data is simulated

Junction 1(H-1)	Junction 2(H-1)	Junction 3(H-1)	Junction 4(H-1)	Junction 1(H)	Junction 2(H)	Junction 3(H)	Junction 4(H)
DateTime							
2015-11-01 01:00:00	0.066225	0.106383	0.044693	0.000000	0.052980	0.106383	0.033520
2015-11-01 02:00:00	0.052980	0.106383	0.033520	0.000000	0.033113	0.085106	0.022346

2015-11-01 03:00:00	0.033113	0.085106	0.022346	0.000000	0.013245	0.106383	0.000000
2015-11-01 04:00:00	0.013245	0.106383	0.000000	0.000000	0.026490	0.127660	0.005587
2015-11-01 05:00:00	0.026490	0.127660	0.005587	0.000000	0.006623	0.021277	0.005587
...	...	...	...	...	...	...	...
2017-06-30 22:00:00	0.562914	0.638298	0.150838	0.444444	0.523179	0.595745	0.139665
2017-06-30 23:00:00	0.523179	0.595745	0.139665	0.611111	0.483444	0.553191	0.212291

Command here used: $X_train = Xy_train[Xy_train.index < '2017-04-01'].iloc[:,0:4]$

Table 6. The table contains the training model's first 4 rows and columns and the testing model's first 4 rows and columns[14]

Junction 1(H-1)	Junction 2(H-1)	Junction 3(H-1)	Junction 4(H-1)
DateTime			
2015-11-01 01:00:00	0.066225	0.106383	0.044693
2015-11-01 02:00:00	0.052980	0.106383	0.033520
2015-11-01 03:00:00	0.033113	0.085106	0.022346
2015-11-01 04:00:00	0.013245	0.106383	0.000000
2015-11-01 05:00:00	0.026490	0.127660	0.005587
...	...	...	...
2017-03-31 19:00:00	0.476821	0.574468	0.178771
2017-03-31 20:00:00	0.496689	0.531915	0.156425
2017-03-31 21:00:00	0.483444	0.638298	0.156425
2017-03-31 22:00:00	0.403974	0.574468	0.150838
2017-03-31 23:00:00	0.423841	0.553191	0.162011

Command here used:

```
y_train = Xy_train[Xy_train.index < '2017-04-01'].iloc[:,4:]
```

```
y_train
```

```
me = pd.concat([d1,d2],axis = 1,join = 'outer')
```

Table 7. The table contains the combining of the two datasets

	DateTime	Junction	Vehicles	ID	DateTime	Junction	ID
0	2015-11-01 00:00:00	1	15	2015110100 1	2017-07-01 00:00:00	1	2017070100 1
1	2015-11-01 01:00:00	1	13	2015110101 1	2017-07-01 01:00:00	1	2017070101 1
2	2015-11-01 02:00:00	1	10	2015110102 1	2017-07-01 02:00:00	1	2017070102 1
3	2015-11-01 03:00:00	1	7	2015110103 1	2017-07-01 03:00:00	1	2017070103 1
4	2015-11-01 04:00:00	1	9	2015110104 1	2017-07-01 04:00:00	1	2017070104 1
...	...	...	...	...	...	...	...
1180 6	2017-03-06 22:00:00	1	81	2017030622 1	2017-10-31 22:00:00	4	2017103122 4
1180 7	2017-03-06 23:00:00	1	72	2017030623 1	2017-10-31 23:00:00	4	2017103123 4

5.2 Results from the simulation

The results of the traffic are as follows which by the matplotlib library[15].

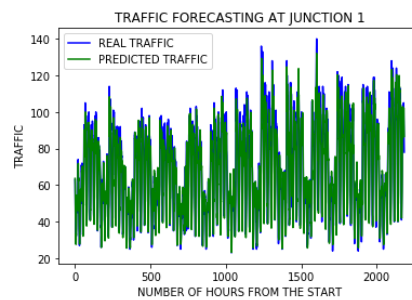


Figure 7. The figure signifies the Traffic prediction of Junction 1 from the datasets

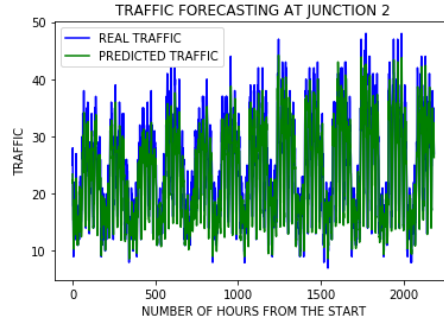


Figure 8. The figure signifies the Traffic prediction of Junction 2 from the datasets

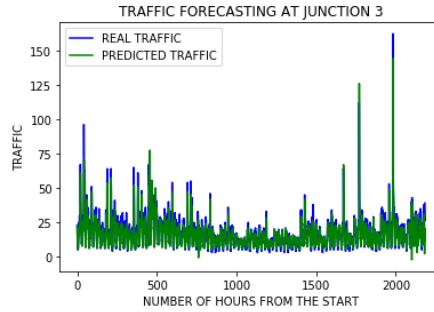


Figure 9. The figure signifies the Traffic prediction of Junction 3 from the datasets

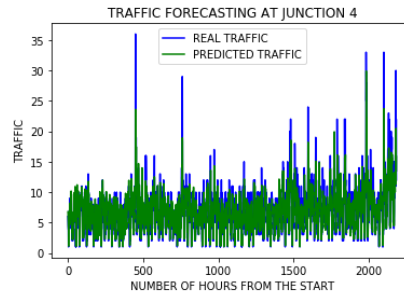


Figure 10. The figure signifies the Traffic prediction of Junction 4 from the datasets

In the last, researchers are suggested to go through various uses of Machine Learning in many useful applications like churn prediction, healthcare, agriculture, Transportation, etc., in [21-32].

6 Conclusion

In the system, it has been concluded that we develop the traffic flow prediction system by using a machine learning algorithm. By using regression model, the prediction is done. The public gets the benefits such as the current situation of

the traffic flow, they can also check what will be the flow of traffic on the right after one hour of the situation and they can also know how the roads are as they can know mean of the vehicles passing through a particular junction that is 4 here. The weather conditions have been changing from years to years. The cost of fuel is also playing a major role in the transportation system. Many people are not able to afford the vehicle because of the fuel cost. So, there can be many variations in the traffic data. There is one more scenario where people prefer going on their own vehicle without car pooling, this also matters in the traffic congestion. So, this prediction can help judging the traffic flow by comparing them with these 2 years data sets. The forecasting or the prediction can help people or the users in judging the road traffic easier before hand and even they can decide which way to go using their navigator and also this will prediction will be also helpful.

7 Future Work

In the future, the system are often further improved using more factors that affect traffic management using other methods like deep learning, artificial neural network, and even big data. The users can then use this technique to seek out which route would be easiest to achieve on destination. The system can help in suggesting the users with their choice of search and also it can help to find the simplest choice where traffic isn't in any crowded environment. Many forecasting methods have already been applied in road traffic jam forecasting. While there's more scope to create the congestion prediction more precise, there are more methods that give precise and accurate results from the prediction. Also, during this period, the employment of the increased available traffic data by applying the newly developed forecasting models can improve the prediction accuracy. These days, traffic prediction is extremely necessary for pretty much every a part of the state and also worldwide. So, this method of prediction would be helpful in predicting the traffic before and beforehand. For better congestion prediction, the grade and accuracy are prominent in traffic prediction. within the future, the expectation are going to be the estimation of established order accuracy prediction with much easier and user-friendly methods so people would find the prediction model useful and that they won't be wasting their time and energy to predict the information. There will be some more accessibility like weather outlook, GPS that's the road and accident-prone areas will be highlighted in order that people wouldn't prefer using the paths which aren't safe and simultaneously they'll predict the traffic. This will be done by deep learning, big data, and artificial neural networks[18].

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